CHAPTER V

CONCLUSIONS AND SUGGESTIONS

5.1 Conclusions

This research set out to investigate a structural blind spot in the deployment of Explainable Artificial Intelligence within multi-tenant Enterprise Resource Planning systems: the hallucinated explanation phenomenon, in which post-hoc explainers continue to produce confidently formatted, visually emphatic outputs even when applied to models trained on data-sparse tenants whose underlying predictive signal has collapsed. The empirical evidence gathered in this study demonstrates that data sparsity degrades explanation fidelity in a non-linear and qualitatively misleading manner. Across the five simulated tenant tiers, and under a strict leakage-avoidance regime in which only order-time observables were admitted into the feature matrix, SHAP rank correlation against the data-rich baseline fell monotonically from 1.0000 to 0.4485 and crossed below the conventional reliability bound of 0.70 at the n = 1,000 tier, while SHAP absolute variance simultaneously rose by more than two orders of magnitude from 0.0224 at the data-rich tier to 5.0392 at the cold-start tier. By contrast, classical predictive metrics did not reveal this regime change: AUC-ROC remained uniformly modest across tiers A--D (0.61 --0.66) and anomalously rose to 0.8889 at the cold-start tier as a small-sample artefact. Taken together, these findings establish that XAI applied to cold-start tenants is not merely less accurate it is unfaithful in rank order and volatile in attribution magnitude, while remaining indistinguishable from a reliable explanation in its rendered visual form. This directly answers Research Question 1.

The study further establishes that the operational reliability of local post-hoc explanations exhibits a regime change an empirical Reliability Cliff at the n = 1,000-record threshold. This boundary is not the product of a single arbitrary metric; rather, it is supported by the simultaneous convergence of two independent explanation-

fidelity indicators at the same tier: the Spearman rank-correlation criterion ($\rho_C = 0.6429 < 0.70$) and the SHAP variance criterion ($\sigma^2|SHAP|,C$ rising 128% above the data-rich baseline). The classical AUC criterion does not provide additional support in the leakage-free regime predictive performance is uniformly weak across tiers A--D and anomalously strong at the cold-start tier which makes the explanation-fidelity criteria not merely convergent with, but operationally necessary for, the localisation of the cliff. Below this threshold, the underlying machine-learning models and their post-hoc explanations diverge from any genuine business logic and cannot be trusted for operational decision-making. The cross-task replication on the Employee Attrition and Lead Scoring tasks (Section 4.6) establishes that the qualitative regime change is not an artefact of the Late Delivery Risk substrate: the SHAP-variance expansion and Spearman degradation recur on both additional ERP-native tasks, while the absolute threshold τ at which the cliff occurs is task-specific. This directly answers Research Question 2.

The study concludes that the conventional practice of deploying a uniform machine-learning and XAI pipeline across every tenant of a multi-tenant SaaS platform is fundamentally flawed, because it ignores the per-tenant data-maturity distribution that is intrinsic to the SaaS business model. The Maturity-Aware AI Routing pattern proposed and implemented in this research provides the necessary architectural correction. By placing the predict_or_heuristic() gate inside the LatePaymentPredictor service of the studied Django ERP, and by parameterising it on the empirically-derived COLD_START_INVOICE_THRESHOLD = 1000, the system structurally protects cold-start tenants from being shown hallucinated local explanations, serving them platform-wide Global Heuristics drawn from data-rich SHAP analyses instead, while continuing to deliver granular local SHAP and LIME explanations to data-mature tenants for whom the empirical evidence supports their reliability. This directly answers Research Question 3.

The principal academic takeaway of this thesis is therefore that Explainable Artificial Intelligence must not be conceived of as a passive reporting feature appended

to a model after the fact, but as an active diagnostic instrument that exposes the limits of model reliability and conditions the way intelligence is delivered to the end user. In multi-tenant SaaS platforms, where data maturity is heterogeneous by construction, this reframing is not optional: it fundamentally shifts the responsibility of the system architect from displaying explanations to governing them, and it establishes data maturity as a first-class routing dimension in any production AI subsystem.

5.2 Suggestions

The findings of this study open several productive directions for further work, which may be organized into suggestions for future academic research and suggestions for further system development.

5.2.1 For Future Research (Academic)

- Future studies should investigate XAI fidelity degradation in longitudinal and live SaaS environments, rather than the cross-sectional simulated datasets used in this thesis. A longitudinal design would allow the interaction between data sparsity and concept drift to be characterised, addressing the realistic case in which a tenant's data distribution itself evolves as the tenant matures.
- The cold-start XAI problem should be extended beyond the tabular gradient-boosting setting studied here to deep learning model families, in particular recurrent and attention-based architectures (e.g., LSTMs and Transformers) applied to time-series ERP forecasting, together with their attendant explainers (Integrated Gradients, Attention rollout). The quantitative threshold derived in this thesis is unlikely to transfer literally to these settings and would need to be re-established empirically.
- Controlled user studies should be conducted with business operators of multi-tenant ERPs in order to measure the perceived trust and decision-making accuracy induced by Global Heuristics versus Local Explanations in cold-start scenarios. This would complement the present technical evidence of explanation unfaithfulness with behavioural evidence of its downstream consequences on end-user judgement.

5.2.2 For System Development (Practical)

- The static COLD_START_INVOICE_THRESHOLD = 1000 constant should be replaced, in production deployments, by a dynamic threshold mechanism that auto-adjusts in response to real-time data-distribution indicators for example, Kolmogorov--Smirnov drift statistics computed against the global reference distribution, or per-tenant data-velocity (records-per-week) metrics. This would allow the routing gate to remain calibrated as both the platform and the individual tenant evolve.
- The Maturity-Aware AI Routing pattern should be generalized beyond the Late Delivery Risk task to the other AI-supported modules of the ERP (Lead Scoring, Employee Attrition, Procurement Risk, Expense-Claim Anomaly). Each module is expected to exhibit a distinct empirical threshold reflecting its task complexity and class prevalence, and the routing layer should accept per-module configuration accordingly.
- An interactive Data Maturity Dashboard should be developed for ERP administrators, visually communicating each tenant's progress toward the data-maturity threshold required for personalised local AI predictions. Such a dashboard converts the architectural threshold from an invisible engineering constant into a transparent business-facing artefact, supports onboarding planning, and aligns end-user expectations with the technical preconditions of trustworthy local explanations.